

Name: _____

GCSE (1 - 9)

Writing an Expression

Solutions:



These questions are compiled from *released* [Pearson Edexcel GCSE maths exam papers](#)

They contain exam questions from up until the **Summer 2023** exam series

More recent exam papers do exist but are withheld for mock exams

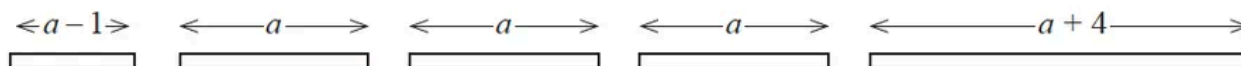
As new exam papers are released these worksheets will gradually be updated

The **most current** version of these worksheets will always be available at: kennethstafford.co.uk

Question	Where the question appeared
1	Edexcel, GCSE, Autumn 2017, Paper 1F, Q6
2	Edexcel, GCSE, Autumn 2017, Paper 1F, Q24
3	Edexcel, GCSE, Autumn 2019, Paper 1F, Q28
4	Edexcel, GCSE, Autumn 2019, Paper 2F, Q18
5	Edexcel, GCSE, Summer 2017, Paper 3F, Q2
6	Edexcel, GCSE, Summer 2019, Paper 1F, Q13
7	Edexcel, GCSE, Summer 2019, Paper 2F, Q7
8	Edexcel, GCSE, Summer 2023, Paper 1F, Q18
9	Edexcel, GCSE, November 2022, Paper 2F, Q14

1

Here are five straight rods.



All measurements are in centimetres.

The total length of the five rods is L cm.

Find a formula for L in terms of a .

Write your formula as simply as possible.

(Total for Question 6 is 3 marks)

[Edexcel, GCSE, Autumn 2017, Paper 1F, Q6]

2

- Kiaria is 7 years older than Jay.
- Martha is twice as old as Kiaria.
- The sum of their three ages is 77

Find the ratio of Jay's age to Kiaria's age to Martha's age.

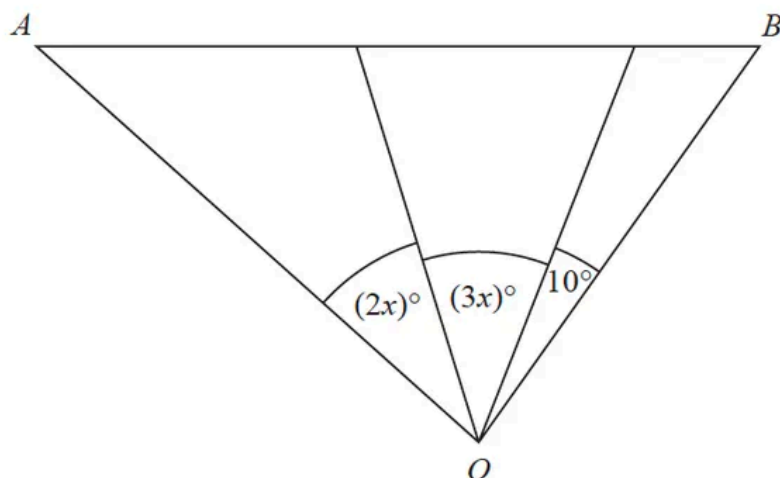
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(Total for Question 24 is 4 marks)

[Edexcel, GCSE, Autumn 2017, Paper 1F, Q24]

3

The diagram shows triangle AOB .



Angle AOB is **not** an obtuse angle.

Find the greatest value of x .

You must show all your working.

(Total for Question 28 is 3 marks)

[Edexcel, GCSE, Autumn 2019, Paper 1F, Q28]

4

Ben is n years old.

Chloe is twice as old as Ben.

Dan is five years younger than Ben.

The total of Ben's age, Chloe's age and Dan's age is T years.

(a) Find a formula for T in terms of n .

(3)

(b) In the table below, put a tick (✓) in the box next to the identity.

$3h + 2 = 14$	
$3a + 4b - 2c$	
$A = \pi r^2$	
$5m - 3m = 2m$	
$x + 7 \leq 12$	

(1)

(Total for Question 18 is 4 marks)

[Edexcel, GCSE, Autumn 2019, Paper 2F, Q18]

5

Cups are sold in packs and in boxes.

There are 12 cups in each pack.

There are 18 cups in each box.

Alison buys p packs of cups and b boxes of cups.

Write down an expression, in terms of p and b , for the total number of cups Alison buys.

(Total for Question 2 is 2 marks)

[Edexcel, GCSE, Summer 2017, Paper 3F, Q2]

6

The length of a line is x centimetres.

Write down an expression, in terms of x , for the length of the line in millimetres.

(Total for Question 13 is 1 mark)

[Edexcel, GCSE, Summer 2019, Paper 1F, Q13]

7

There are y boats on a lake.

There are 7 people in each boat.

Write an expression, in terms of y , for the total number of people in the boats.

(Total for Question 7 is 1 mark)

[Edexcel, GCSE, Summer 2019, Paper 2F, Q7]

8

Naomi has b bags of apples and c crates of apples.

There are 5 apples in each bag.

There are 28 apples in each crate.

Naomi has a total of T apples.

Write a formula for T in terms of b and c .

(Total for Question 18 is 3 marks)

[Edexcel, GCSE, Summer 2023, Paper 1F, Q18]

9

There are x sweets in a box.

There are y sweets in a packet.

- (c) Write an expression, in terms of x and y , for the total number of sweets in 3 boxes and 2 packets.

.....
(2)

(Total for Question 14 is 7 marks)

[Edexcel, GCSE, November 2022, Paper 2F, Q14]